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- 1) Compare the two technology scaling methods, namely, (i) the Constant electric field scaling and (ii) the Constant power supply Voltage scaling. In particular, show analytically by using equations how the delay time, power dissipation, and power density are affected in terms of scaling factor s . To be more specific, what would happen if the design rule changes from say, $1\mu\text{m}$ to $1/s\mu\text{m}$ ($s > 1$)?

301:

Technology Scaling means reducing the size of transistor to improve speed, density and performance.

1, Constant electric field:

All dimension and Voltages are reduced proportionally so that electric field remain constant inside the transistor.

- Delay time : $\text{Delay} \propto \frac{1}{s}$
- Power Dissipation : $\text{power} \propto \frac{1}{s^2}$
- Power Density = Constant.

2, Constant power supply Voltage Scaling :

only transistor dimension are reduced, while the supply voltage is kept constant

- Delay time $\rightarrow \text{Delay} \propto \frac{1}{s}$
- Power Dissipation $\rightarrow \text{power} \propto \frac{1}{s}$
- Power Density $\propto s$

Comparison

parameter	Constant E -field	Constant Voltage supply
Supply Voltage	Reduced	Constant
Electric field	Constant	increase
Delay	Decrease	Decrease
power Dissipation	Decrease	Decrease
power Density	Constant	increase
Heat	No	Yes.